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Unmanned aerial vehicle base station and unmanned aerial vehicle system

Abstract

The embodiments of the present disclosure involve the technical field of unmanned aerial vehicles, and disclose an unmanned aerial vehicle base station and an unmanned aerial vehicle system. The unmanned aerial vehicle base station includes a bracket, a tarmac, a charging component, and a temperature adjusting component. The bracket is provided with an accommodating cavity, the tarmac separates the accommodating cavity into an upper compartment chamber and a lower compartment chamber, the temperature adjusting component includes a semiconductor module, a first ventilation assembly, a second ventilation assembly, and a third ventilation assembly, and the semiconductor module is partially provided on the first ventilation assembly and partially provided on the second ventilation assembly.

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Background/Summary

CROSS REFERENCE TO RELATED DISCLOSURE

(1) This patent claims priority of Chinese Patent Application No. 2023101066350.9, filed Feb. 3, 2023, the entirety of which is hereby incorporated herein by reference.

TECHNICAL FIELD

(2) The embodiments of the present disclosure relate to the technical field of unmanned aerial vehicles, particularly to an unmanned aerial vehicle base station and an unmanned aerial vehicle system.

BACKGROUND

(3) Unmanned aerial vehicles can be used to perform aerial photography, drug spraying, explorations, or like tasks because of their flexible operations. As one of the supporting facilities of unmanned aerial vehicles, an unmanned aerial vehicle base station is generally used to provide at least docking service, cleaning service, charging service, and like services for unmanned aerial vehicles when the unmanned aerial vehicle does not perform tasks.

(4) In implementing the embodiments of the present disclosure, the inventors have found that: the existing heat dissipation systems of unmanned aerial vehicle base stations are single and simple,

and most of the unmanned aerial vehicle base stations use fans directly to dissipate heat. When the unmanned aerial vehicle is docked at the unmanned aerial vehicle base station for charging, the temperatures of the unmanned aerial vehicle and the unmanned aerial vehicle base station are relatively high, and a simple heat dissipation system cannot effectively reduce the temperatures of the unmanned aerial vehicle and the base station, which reduces the charging efficiency of the unmanned aerial vehicle by the unmanned aerial vehicle base station.

SUMMARY

(5) The technical problems to be mainly solved by the embodiments of the present disclosure are to provide an unmanned aerial vehicle base station and an unmanned aerial vehicle system that can perform heat dissipation on multiple areas of the unmanned aerial vehicle base station in a targeted manner, and effectively improve the working performance of the unmanned aerial vehicle base station.

(6) The first aspect of the present disclosure is to provide an unmanned aerial vehicle base station, comprises a bracket, a tarmac, a charging component, and a temperature adjusting component. The bracket is provided with an accommodating cavity; the tarmac separates the accommodating cavity into an upper compartment chamber and a lower compartment chamber, the tarmac being configured to dock an unmanned aerial vehicle; the charging component is configured to provide electrical energy to an unmanned aerial vehicle docked at the tarmac; and a temperature adjusting component is provided in the lower compartment chamber, wherein the temperature adjusting component includes a semiconductor module, a first ventilation assembly, a second ventilation assembly, and a third ventilation assembly, and the semiconductor module is partially provided in the first ventilation assembly and partially provided in the second ventilation assembly; wherein the first ventilation assembly is configured to perform heat dissipation on the upper compartment chamber; the second ventilation assembly is configured to perform heat dissipation on part of the semiconductor module; and the third ventilation assembly is configured to perform heat dissipation on the lower compartment chamber.

(7) The second aspect of this disclosure is to provide an unmanned aerial vehicle system, including an unmanned aerial vehicle and the unmanned aerial vehicle base station as described above.

(8) An unmanned aerial vehicle base station in an embodiment of the present disclosure includes a bracket, a tarmac, a charging mechanism, and a temperature adjusting mechanism. The bracket includes an upper compartment cover and a lower compartment cover, and the tarmac is configured to dock an unmanned aerial vehicle. The temperature adjusting mechanism includes a semiconductor module, a first ventilation assembly, a second ventilation assembly, and a third ventilation assembly. An upper end of the semiconductor module is provided in the first ventilation assembly, and a lower end of the semiconductor module is provided in the second ventilation assembly; and components and parts such as the charging mechanism are provided in the third ventilation assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to state specific embodiments of the present disclosure or the technical solutions in the prior art more clearly, the following will briefly introduce the drawings needed in the description of specific embodiments and the prior art. Throughout the drawings, like elements or portions are generally identified by like reference numerals. In the drawings, elements or portions may not necessarily be drawn to scale.

(2) FIG. 1 is an exploded view of an unmanned aerial vehicle base station from one viewing angle according to an embodiment of the present disclosure.

(3) FIG. 2 is an exploded view of an unmanned aerial vehicle base station from another viewing

angle according to an embodiment of the present disclosure.

(4) FIG. 3 is a schematic view of a lower compartment cover of an unmanned aerial vehicle base station according to an embodiment of the present disclosure.

(5) FIG. 4 is an exploded view of a temperature adjusting mechanism of an unmanned aerial vehicle base station from one viewing angle according to an embodiment of the present disclosure.

(6) FIG. 5 is a sectional view of a temperature adjusting mechanism of an unmanned aerial vehicle base station from one viewing angle according to an embodiment of the present disclosure.

(7) FIG. 6 is a schematic view of a temperature adjusting mechanism of an unmanned aerial vehicle base station from one viewing angle according to an embodiment of the present disclosure.

(8) FIG. 7 is a schematic view of an unmanned aerial vehicle system from one viewing angle according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

(9) In order to make the present disclosure readily understood, a more particular description of the disclosure will be rendered by reference to accompanying drawings and specific embodiments. It needs to be noted that when an element is referred to as being “secured” to another element, it can be directly on another element or one or more intervening elements may be present in between.

When one element is referred to as being “connected” to another element, it can be directly connected to another element or one or more intervening elements may be present in between. As used in the description, the orientations or positional relationships indicated by the terms “up”, “down”, “inner”, “outer”, “vertical”, “horizontal” and the like are based on the orientations or positional relationships shown in the drawings for purposes of describing the disclosure and simplifying the description only, and are not intended to indicate or imply that the referenced device or element must have a particular orientation or be constructed and operated in a particular orientation. It is therefore not to be understood as limiting the disclosure. Furthermore, the terms “first”, “second”, and the like are used for descriptive purposes only and are not to be construed as indicating or implying relative importance.

(10) Unless defined otherwise, all technical and scientific terms used in the specification have the same meaning as commonly understood by one of ordinary skill in the art to which the present disclosure belongs. The terminology used in the description of the present disclosure is for the purpose of describing specific embodiments only and is not intended to limit the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(11) Further, the technical features involved in different embodiments of the present disclosure described below can be combined with each other as long as they do not conflict with each other.

(12) Referring to FIGS. 1 to 3, an unmanned aerial vehicle base station **100** includes a bracket **10**, a tarmac **20**, a temperature adjusting component **30**, and a charging component **40**. The bracket **10** is provided with an accommodating cavity, the tarmac **20** is provided in the accommodating cavity, and the tarmac **20** is configured to the unmanned aerial vehicle to dock. The charging component **40** is provided in the accommodating cavity, and the charging component **40** is configured to supplying electric energy to the unmanned aerial vehicle docked on the tarmac **20**. The temperature adjusting component **30** is provided in the accommodating cavity, and the temperature adjusting component **30** is configured to adjust the temperature in the accommodating cavity so that the temperature in the accommodating cavity is maintained within a temperature range suitable for charging an unmanned aerial vehicle, thereby ensuring the charging efficiency of the unmanned aerial vehicle by the unmanned aerial vehicle base station **100**.

(13) The bracket **10** comprises an upper compartment cover **11** and a lower compartment cover **12**. When the upper compartment cover **11** and the lower compartment cover **12** are closed, a closed accommodating cavity is formed. Further, the upper compartment cover **11** is provided with an upper compartment chamber **111**, the lower compartment cover **12** is provided with a lower compartment chamber **121**, the tarmac **20** is provided on the lower compartment cover **12**, the

upper compartment chamber **111** is configured to accommodate the unmanned aerial vehicle, and the lower compartment chamber **121** is configured to accommodate the temperature adjusting component **30** and the charging component **40**.

(14) In some embodiments, the upper compartment cover **11** is arc-shaped towards the inner wall surface of the lower compartment cover **12**. When the upper compartment cover **11** is closed on the lower compartment cover **12** to form a closed accommodating cavity, the inner wall surface of the arc-shaped structure of the upper compartment cover **11** is more conducive to the airflow stream formed when the temperature adjusting component **30** adjusts the temperature of the upper compartment chamber **111**, effectively avoiding the phenomenon of airflow disturbance.

(15) Referring to FIG. 2, the unmanned aerial vehicle base station **100** further comprises a first thermal insulation layer **41**. The first thermal insulation layer **41** is provided on the inner wall surface of the upper compartment cover **11**, and the first thermal insulation layer **41** is configured to reduce the heat exchange efficiency of the upper compartment chamber **111** with the outside. As an example, the first thermal insulation layer **41** may be an insulating cotton, a ceramic fiber layer, or the like.

(16) Referring to FIG. 3, the lower compartment cover **12** is provided with a first ventilation opening **122** and a second ventilation opening **123** that communicate the lower compartment chamber **121** with the outside. Both the first ventilation opening **122** and the second ventilation opening **123** are configured to communicate with the temperature adjusting component **30** so that the temperature adjusting component **30** performs heat dissipation on the lower compartment chamber **121**.

(17) It could be understood that since the unmanned aerial vehicle base station **100** is generally mounted outdoors for use, in order to prevent external animals and sundries from entering the lower compartment chamber **121** which would otherwise affect the normal operation of the unmanned aerial vehicle base station **100**, the unmanned aerial vehicle base station **100** further includes a first barrier net **42** and a second barrier net **43**. The first barrier net **42** is detachably provided at the first ventilation opening **122**, and the second barrier net **43** is detachably provided at the second ventilation opening **123**.

(18) In some embodiments, referring to FIG. 2, the unmanned aerial vehicle base station **100** also includes a drive assembly **44**. The drive assembly **44** is provided to the bracket **10**, the drive assembly **44** being configured to drive movement of the upper compartment cover **11** relative to the lower compartment cover **12** to effect opening or closing of the accommodating cavity by the upper compartment cover **11**. The drive assembly **44** is connected to a controller to further enhance the intelligence of the unmanned aerial vehicle base station **100**.

(19) It could be appreciated that in order to reduce the space occupied by the upper compartment cover **11** after it is opened, the outer wall surface of the lower compartment cover **12** is arc-shaped, and at least part of the upper compartment cover **11** is nested with at least part of the lower compartment cover **12** when the drive assembly **44** drives the upper compartment cover **11** to open the upper compartment chamber **111**.

(20) Referring to FIG. 3, the unmanned aerial vehicle base station **100** further includes a first sealing assembly **45** in order to further improve the sealing property after the upper compartment cover **11** and the lower compartment cover **12** are closed. The first sealing assembly **45** for sealing the gap between the upper compartment cover **11** and the lower compartment cover **12** includes a first sealing member **451** and a second sealing member **452**. The first sealing member **451** is provided on one side of a connecting end face of the lower compartment cover **12** facing the upper compartment cover **11**, and the second sealing member **452** is provided on one side of a connecting end face of the lower compartment cover **12** facing away from the upper compartment cover **11**. By the above arrangement, it is possible to reduce the gap after the upper compartment cover **11** and the lower compartment cover **12** are rotationally closed, effectively improving the air tightness of the upper compartment chamber **111**.

(21) With regard to the tarmac **20** described above, referring to FIG. **1**, the tarmac **20** is provided with a first circulation air port **21** and a second circulation air port **22** penetrating through the tarmac **20** so as to make the upper compartment chamber **111** and the lower compartment chamber **121** in airflow communication. The first circulation air port **21** and the second circulation air port **22** both communicate with the temperature adjusting component **30**, and the temperature adjusting component **30** located in the lower compartment chamber **121** performs airflow exchange with the upper compartment chamber **111** via the first circulation air port **21** and the second circulation air port **22**, thereby adjusting the temperature in the upper compartment chamber **111**.

(22) It could be understood that, in other embodiments, the number of the first circulation air ports **21** is at least two, and at least two of the first circulation air ports **21** are provided in an array or randomly. The number of the second circulation air ports **22** is at least two, and at least two of the second circulation air ports **22** are provided in an array or randomly. The number of the first circulation air ports **21** is at least two, so that the aperture of each first circulation air port **21** is made small, and likewise, the number of the second circulation air ports **22** is at least two, so that the aperture of each second circulation air port **22** is made small, for example, 1 mm, 3 mm, 5 mm, etc. so as to effectively block sundries which enter the upper compartment chamber **111** from the outside from continuing to enter the lower compartment chamber **121**, which would otherwise affect the normal adjustment operation of the temperature adjusting component **30**.

(23) With regard to the temperature adjusting component **30** described above, referring to FIGS. **4** to **6**, the temperature adjusting component **30** includes a semiconductor module **31**, a first ventilation assembly, a second ventilation assembly, a third ventilation assembly, and an adapter **35**. A part of the semiconductor module **31** is provided in the first ventilation assembly, and the first ventilation assembly cooperates with the part of the semiconductor module **31** located in the first ventilation assembly to adjust the temperature of the upper compartment chamber **111**. A part of the semiconductor module **31** is provided in the second ventilation assembly, and the second ventilation assembly is used for performing heat dissipation on the part of the semiconductor module **31** located in the second ventilation assembly, so as to avoid the situation that the semiconductor module **31** works at such an excessively high temperature that it affects the working performance. The third ventilation assembly is in communication with the lower compartment chamber **121**, and the third ventilation assembly is used for discharging the heat generated by components and parts such as the charging component **40** during operation to the outside in time, so as to avoid excessive temperature of the lower compartment chamber **121** caused by heat accumulation in the lower compartment chamber **121**. The adapter **35** is provided in the lower compartment chamber **121**, the adapter **35** is electrically connected to the semiconductor module **31**, and the adapter **35** is configured to supply electric energy to the semiconductor module **31**.

(24) The first ventilation assembly includes a first fan **321** and a first air passage **322**. The first air passage **322** is provided with a first opening **3221** and a second opening **3222** in communication with the upper compartment chamber **111**, wherein the first opening **3221** is in airflow communication with the first circulation air port **21** of the tarmac **20**, and the second opening **3222** is in airflow communication with the second circulation air port **22** of the tarmac **20**; the upper end **311** of the semiconductor module **31** is located in the first air passage **322**; and the upper end **311** of the semiconductor module **31** can heat or cool the gas in the first air passage **322**. The first fan **321** is provided at the first opening **3221**, and the first fan **321** is configured to provide airflow power so that a hot airflow or a cold airflow generated in the first air passage **322** is blown into the upper compartment chamber **111**, thereby achieving temperature adjustment of the upper compartment chamber **111**.

(25) The unmanned aerial vehicle base station **100** further includes a second thermal insulation layer **46**, and the second thermal insulation layer **46** is provided on the inner wall surface of the first air passage **322**. Since the air heated or cooled by the upper end **311** of the semiconductor module **31** in the first air passage **322** is supplied into the upper compartment chamber **111** to adjust

the temperature in the upper compartment chamber **111**, providing the second thermal insulation layer **46** on the inner wall surface of the first air passage **322** can reduce the heat exchange between the hot air or cold air with the outside so as to improve the temperature adjustment efficiency of the upper compartment chamber **111**, and the temperature of the upper compartment chamber **111** can be increased or decreased in a relatively short time. By way of example, the second thermal insulation layer **46** may be insulating cotton, a ceramic fiber layer, or the like.

(26) The unmanned aerial vehicle base station **100** further includes a second sealing assembly **47**, wherein the second sealing assembly **47** is provided between the tarmac **20** and the first ventilation assembly, and the second sealing assembly **47** is configured to seal the gap between the first opening **3221** of the first air passage **322** and the first circulation air port **21** of the tarmac **20**, and seal the gap between the second opening **3222** of the first air passage **322** and the second circulation air port **22** of the tarmac **20**, preventing the hot airflow or the cold airflow in the first air passage **322** from leaving the first air passage **322** from the gap, and reducing the temperature adjustment efficiency of the upper compartment chamber **111**, and at the same time, avoiding the noise generated when the airflow in the first air passage **322** flows out from the gap, so as to improve the quality of the unmanned aerial vehicle base station **100**.

(27) The second ventilation assembly includes a second fan **331** and a second air passage **332**. The second air passage **332** is provided with a third opening **3321** and a fourth opening **3322**, wherein the third opening **3321** is in airflow communication with the first ventilation opening **122** of the lower compartment cover **12**, and the fourth opening **3322** is in airflow communication with the second ventilation opening **123** of the lower compartment cover **12**; the lower end **312** of the semiconductor module **31** is located in the second air passage **332**; when the upper end **311** of the semiconductor module **31** is in a cooling state, the lower end **312** of the semiconductor module **31** is in a heating state; and the second fan **331** is configured to discharge the heat generated by the lower end **312** of the semiconductor module **31** to the outside through the second air passage **332** in time. It could be understood that path L1 and path L2 formed by the arrows in FIG. 5 represent the airflow stream in the first air passage **322** and the second air passage **332**, respectively. It is worth noting that the direction of the arrow does not limit the direction of the airflow stream.

(28) The third ventilation assembly includes a third fan **341** and a third air passage **342**. The third air passage **342** is provided with a fifth opening **3421** and a sixth opening **3422** communicating with the lower compartment chamber **121**, wherein the fifth opening **3421** is in airflow communication with the first ventilation opening **122** of the lower compartment cover **12**, and the sixth opening **3422** is in airflow communication with the first ventilation opening **122** of the lower compartment cover **12**. Components and parts such as the charging component **40** and the adapter **35** are provided in the third air passage **342**, and a third fan **341** is provided in the fifth opening **3421**. The third fan **341** is configured to provide airflow power so as to discharge the heat generated by components and parts such as the charging component **40** and the adapter **35** in operation to the outside in time.

(29) In other embodiments, the third ventilation assembly further includes a fourth fan **343** provided at the sixth opening **3422** in order to further improve the fluidity of the airflow in the lower compartment chamber **121** and enhance the heat dissipation effect. The third fan **341** and the fourth fan **343** cooperate together to perform heat dissipation on the components and parts in the lower compartment chamber **121** by means of blowing and sucking. As an example, the third fan **341** is used to suck relatively cold air from the outside into the lower compartment chamber **121**, and relatively warm air after convective heat dissipation is performed on the components and parts such as the charging component **40** and the adapter **35** is blown to the outside by the fourth fan **343** to complete the blowing and sucking process. It could be understood that the path L3 formed by the arrows in FIG. 6 represents the airflow stream in the third air passage **342**. It is worth noting that the direction of the arrow does not restrict the direction of the airflow stream.

(30) Referring to FIG. 1, the temperature adjusting component **30** further includes a first backup

battery **36**. The first backup battery **36** is provided in the lower compartment chamber **121** and located in the third air passage **342**, the first backup battery **36** is electrically connected to the adapter **35** and the semiconductor module **31**, respectively, and the first backup battery **36** is configured to supply electric energy to the semiconductor module **31**. It is worth noting that the direction of current when the adapter **35** directly supplies power to the semiconductor module **31** is opposite to the direction of current when the adapter **35** supplies power to the semiconductor module **31** through the first backup battery **36**, so that the operating states of the upper end **311** and the lower end **312** of the semiconductor module **31** are reversed. As an example, when the adapter **35** directly supplies power to the semiconductor module **31**, for example, the upper end **311** of the semiconductor module **31** is in a cooling state, and the lower end **312** of the semiconductor module **31** is in a heat dissipation state; and when the adapter **35** supplies power to the semiconductor module **31** through the first backup battery **36**, the upper end **311** of the semiconductor module **31** is in a heat radiation state, and the lower end **312** of the semiconductor module **31** is in a cooling state.

(31) In other embodiments, the temperature adjusting component **30** further includes a second backup battery. The second backup battery is respectively electrically connected to an external power source and the adapter **35**, wherein the second backup battery is configured to store electric energy, so as to avoid the situation that the external power source cannot continuously supply power to the adapter **35**, for example, in special cases of power cut, power failure, etc. Therefore, the second backup battery can continue to supply electric energy to the adapter **35**.

(32) An unmanned aerial vehicle base station **100** in an embodiment of the present disclosure includes a bracket **10**, a tarmac **20**, a charging component **40**, a temperature adjusting component **30**, and a first thermal insulation layer **41**. The bracket **10** includes an upper compartment cover **11** and a lower compartment cover **12**, and the tarmac **20** is configured to dock an unmanned aerial vehicle. The temperature adjusting component **30** includes a semiconductor module **31**, a first ventilation assembly, a second ventilation assembly, and a third ventilation assembly. An upper end **311** of the semiconductor module **31** is provided in the first ventilation assembly, and a lower end **312** of the semiconductor module **31** is provided in the second ventilation assembly; and components and parts such as the charging component **40** are provided in the third ventilation assembly. With the above-mentioned structure, the ventilation and heat dissipation channels of the unmanned aerial vehicle base station **100** can have a clear division of labor, targeted heating and heat dissipation can be realized, and the working performance of each component and part can be effectively improved. In addition, providing the first thermal insulation layer **41** on the upper compartment cover **11** can effectively reduce the heat exchange efficiency between the upper compartment cover **11** and the outside, thereby realizing the scenario that the upper compartment cover **11** has the function of thermal insulation.

(33) The present disclosure also provides an embodiment of an unmanned aerial vehicle system **1000**. With reference to FIG. 7, the unmanned aerial vehicle system **1000** includes an unmanned aerial vehicle base station **100** and an unmanned aerial vehicle **200**, wherein the unmanned aerial vehicle base station **100** at least provides docking and charging services for the unmanned aerial vehicle **200**. The structure and function of the unmanned aerial vehicle base station **100** can be referred to the above embodiments, and will not be elaborated here.

(34) Alternatively, the first ventilation assembly includes a first fan and a first air passage, the first air passage is provided with a first opening and a second opening communicating with the upper compartment chamber, the first fan is provided at the first opening, and an upper end of the semiconductor module is located in the first air passage.

(35) Alternatively, the second ventilation assembly includes a second fan and a second air passage, the second air passage is provided with a third opening and a fourth opening communicating with an outside, the second fan is provided at the third opening, and a lower end of the semiconductor module is located in the second air passage.

(36) Alternatively, the temperature adjusting component further includes an adapter, and the adapter is respectively electrically connected to the semiconductor module, the first fan, and the second fan; and when the semiconductor module supplies power through the adapter, the upper end of the semiconductor module is in a cooling state, and the lower end of the semiconductor module is in a heat dissipation state.

(37) Alternatively, the temperature adjusting component further includes a first backup battery, and the first backup battery is respectively electrically connected to the adapter and the semiconductor module; and when the semiconductor module supplies power through the first backup battery, the upper end of the semiconductor module is in a heat dissipation state, and the lower end of the semiconductor module is in a cooling state.

(38) Alternatively, the temperature adjusting component further includes a second backup battery electrically connected to the adapter, the second backup battery is configured to store electrical energy and provide electrical energy to the adapter.

(39) Alternatively, the third ventilation assembly includes a third fan and a third air passage, the third air passage is in communication with the lower compartment chamber, the third air passage is provided with a fifth opening and a sixth opening in communication with the outside, and the third fan is provided at the fifth opening; and the third fan is configured to provide airflow power to perform heat dissipation on the charging component, the adapter, and the first backup battery located in the third air passage.

(40) Alternatively, the third ventilation assembly further includes a fourth fan provided at the sixth opening, the third fan being configured to draw outside air into the third air passage, and the fourth fan being configured to blow air in the third air passage to the outside.

(41) Alternatively, the bracket is provided with a first ventilation opening and a second ventilation opening, and the second ventilation assembly conducts air inflow from the first ventilation opening and air outflow from the second ventilation opening; and the third ventilation assembly conducts air inflow from the first ventilation opening and air outflow from the second ventilation opening; and the unmanned aerial vehicle base station further includes a first barrier net and a second barrier net, wherein the first barrier net is provided at the first ventilation opening, and the second barrier net is provided at the second ventilation opening.

(42) The above-mentioned description is merely an embodiment of the present disclosure, and does not limit the scope of the present disclosure. Any equivalent structure or equivalent process change made by using the contents of the description and the drawings of the present disclosure or directly or indirectly used in other relevant technical fields is likewise included in the scope of protection of the present disclosure.

Claims

1. An unmanned aerial vehicle base station, comprising: a bracket provided with an accommodating cavity; a tarmac separating the accommodating cavity into an upper compartment chamber and a lower compartment chamber, the tarmac is configured to dock an unmanned aerial vehicle; a charging component configured to provide electrical energy to the unmanned aerial vehicle docked at the tarmac; and a temperature adjusting component provided in the lower compartment chamber, wherein the temperature adjusting component comprises a semiconductor module, a first ventilation assembly, a second ventilation assembly, and a third ventilation assembly, and the semiconductor module is partially provided in the first ventilation assembly and partially provided in the second ventilation assembly; wherein the first ventilation assembly is configured to perform heat dissipation on the upper compartment chamber; the second ventilation assembly is configured to perform heat dissipation on part of the semiconductor module; and the third ventilation assembly is configured to perform heat dissipation on the lower compartment chamber.

2. The unmanned aerial vehicle base station according to claim 1, wherein the first ventilation assembly comprises a first fan and a first air passage, the first air passage is provided with a first opening and a second opening communicating with the upper compartment chamber, the first fan is disposed at the first opening, and an upper end of the semiconductor module is disposed in the first air passage.
3. The unmanned aerial vehicle base station according to claim 2, wherein the second ventilation assembly comprises a second fan and a second air passage, the second air passage is provided with a third opening and a fourth opening communicating with an outside, the second fan is disposed at the third opening, and a lower end of the semiconductor module is disposed in the second air passage.
4. The unmanned aerial vehicle base station according to claim 3, wherein the temperature adjusting component further comprises an adapter, and the adapter is respectively electrically connected to the semiconductor module, the first fan, and the second fan; and when the semiconductor module supplies power through the adapter, the upper end of the semiconductor module is in a cooling state, and the lower end of the semiconductor module is in a heat dissipation state.
5. The unmanned aerial vehicle base station according to claim 4, wherein the temperature adjusting component further comprises a first backup battery, and the first backup battery is respectively electrically connected to the adapter and the semiconductor module; and when the semiconductor module supplies power through the first backup battery, the upper end of the semiconductor module is in the heat dissipation state, and the lower end of the semiconductor module is in the cooling state.
6. The unmanned aerial vehicle base station according to claim 4, wherein the temperature adjusting component further comprises a second backup battery electrically connected to the adapter, the second backup battery is configured to store electrical energy and provide electrical energy to the adapter.
7. The unmanned aerial vehicle base station according to claim 5, wherein the third ventilation assembly comprises a third fan and a third air passage, the third air passage is in communication with the lower compartment chamber, the third air passage is provided with a fifth opening and a sixth opening in communication with the outside, and the third fan is disposed at the fifth opening; and the third fan is configured to provide airflow power to perform heat dissipation on the charging component, the adapter, and the first backup battery located in the third air passage.
8. The unmanned aerial vehicle base station according to claim 7, wherein the third ventilation assembly further comprises a fourth fan provided at the sixth opening, the third fan being configured to draw outside air into the third air passage, and the fourth fan is configured to blow air in the third air passage to the outside.
9. The unmanned aerial vehicle base station according to claim 1, wherein the bracket is provided with a first ventilation opening and a second ventilation opening, and the second ventilation assembly conducts air inflow from the first ventilation opening and air outflow from the second ventilation opening; and the third ventilation assembly conducts air inflow from the first ventilation opening and air outflow from the second ventilation opening; and the unmanned aerial vehicle base station further comprises a first barrier net and a second barrier net, wherein the first barrier net is provided at the first ventilation opening, and the second barrier net is provided at the second ventilation opening.
10. The unmanned aerial vehicle base station according to claim 1, wherein the bracket further comprises a upper compartment cover and a lower compartment cover; the unmanned aerial vehicle base station further comprises a first thermal insulation layer is positioned on a inner wall of the upper compartment cover, and the first thermal insulation layer is configured to reduce the efficiency of heat exchange between the upper compartment chamber and an external environment.
11. The unmanned aerial vehicle base station according to claim 10, wherein the unmanned aerial

vehicle base station further comprises a first sealing component, and the first sealing component is configured to seal a first gap between the upper compartment cover and the lower compartment cover.

12. The unmanned aerial vehicle base station according to claim 11, wherein the first sealing component comprising a first sealing element and a second sealing element, the first sealing element is positioned on a connecting end of the lower compartment cover facing towards one side of the upper compartment cover, and the second sealing element is positioned on the connecting end of the lower compartment cover facing away from one side of the upper compartment cover.

13. The unmanned aerial vehicle base station according to claim 2, wherein the unmanned aerial vehicle base station further comprises a second thermal insulation layer, and the second thermal insulation layer is positioned on a inner wall of the first air passage.

14. The unmanned aerial vehicle base station according to claim 3, wherein the unmanned aerial vehicle base station further comprises a second sealing component, and the second sealing component is configured to seal a second gap between the first air passage and a first circulation vent and a second circulation vent.

15. The unmanned aerial vehicle base station according to claim 13, wherein the inner wall surface of the upper compartment cover facing towards the lower compartment cover being curved.

16. The unmanned aerial vehicle base station according to claim 10, wherein the unmanned aerial vehicle base station further comprises comprising a driving component, and the driving component is positioned on the bracket, the driving component is configured to drive the rotation of the upper compartment cover to open or close the upper compartment chamber.

17. An unmanned aerial vehicle system, comprising an unmanned aerial vehicle and an unmanned aerial vehicle base station, the unmanned aerial vehicle base station is configured to dock the unmanned aerial vehicle, wherein the unmanned aerial vehicle base station comprises: a bracket provided with an accommodating cavity; a tarmac separating the accommodating cavity into an upper compartment chamber and a lower compartment chamber, the tarmac is configured to dock an unmanned aerial vehicle; a charging component configured to provide electrical energy to the unmanned aerial vehicle docked at the tarmac; and a temperature adjusting component provided in the lower compartment chamber, wherein the temperature adjusting component comprises a semiconductor module, a first ventilation assembly, a second ventilation assembly, and a third ventilation assembly, and the semiconductor module is partially provided in the first ventilation assembly and partially provided in the second ventilation assembly; wherein the first ventilation assembly is configured to perform heat dissipation on the upper compartment chamber; the second ventilation assembly is configured to perform heat dissipation on part of the semiconductor module; and the third ventilation assembly is configured to perform heat dissipation on the lower compartment chamber.

18. The unmanned aerial vehicle system according to claim 17, wherein the bracket further comprises a upper compartment cover and a lower compartment cover; the unmanned aerial vehicle base station further comprises a first thermal insulation layer is positioned on a inner wall of the upper compartment cover, and the first thermal insulation layer is configured to reduce the efficiency of heat exchange between the upper compartment chamber and an external environment.

19. The unmanned aerial vehicle system according to claim 18, wherein wherein the unmanned aerial vehicle base station further comprises comprising a driving component, and the driving component is positioned on the bracket, the driving component is configured to drive the rotation of the upper compartment cover to open or close the upper compartment chamber.
